Improved NP-Hardness of Approximation for Max-3-Cut and for Max-k-Cut with Small k

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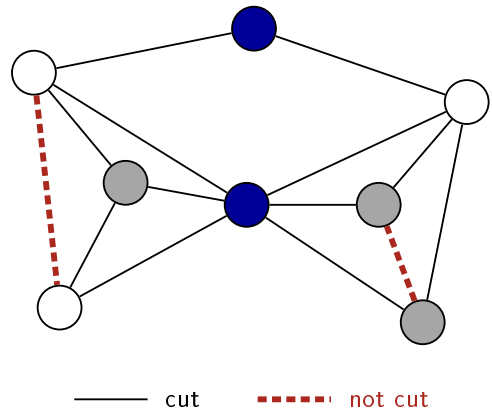
STOC 2027 submission video

The problem: Max-k-Cut

Input: graph with non-negative edge weights

Colour every vertex with one of k colours

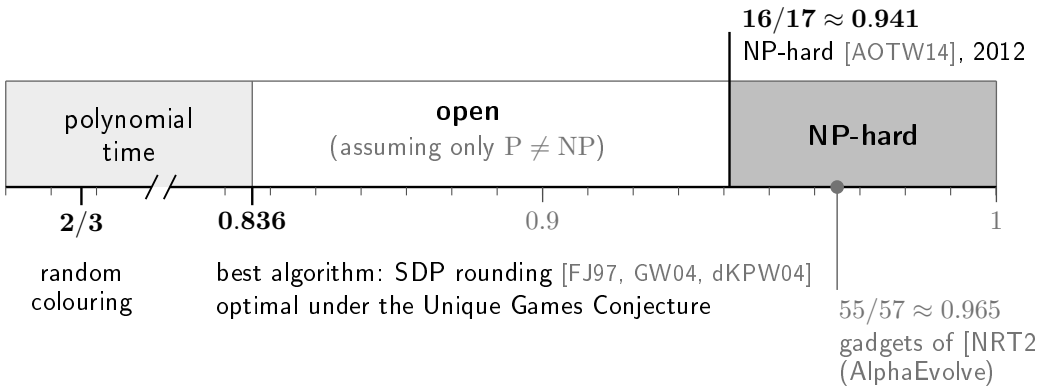
Maximise the total weight of **cut** edges: edges whose endpoints get different colours



- $k = 2$: Max-Cut
- random colouring: each edge is cut with probability $1 - \frac{1}{k}$ ($\frac{2}{3}$ for $k = 3$)

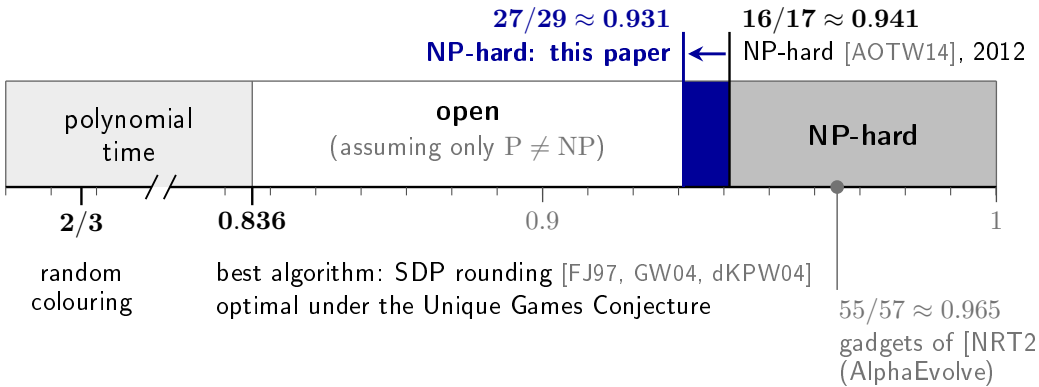
Max-3-Cut: what was known

Which approximation ratios can be achieved in polynomial time?



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Which approximation ratios can be achieved in polynomial time?



Main results

Theorem 1.1 (Max-3-Cut)

For every $\varepsilon > 0$ it is NP-hard to distinguish

- **some** 3-colouring cuts $\geq \frac{29}{30} - \varepsilon$
- **every** 3-colouring cuts $\leq \frac{9}{10} + \varepsilon$

(as fractions of the total edge weight)

$\Longrightarrow$ NP-hard to approximate within

$$\mathbf{27/29 + \varepsilon \approx 0.931}$$

previous: $16/17 \approx 0.941$ [AOTW14]

Theorem 1.2 (every $k \geq 3$)

NP-hard to approximate within

$1 - 2/\Gamma_{k-3} + \varepsilon, \quad \Gamma_L = 29 + 18L + 3\binom{L}{2}$

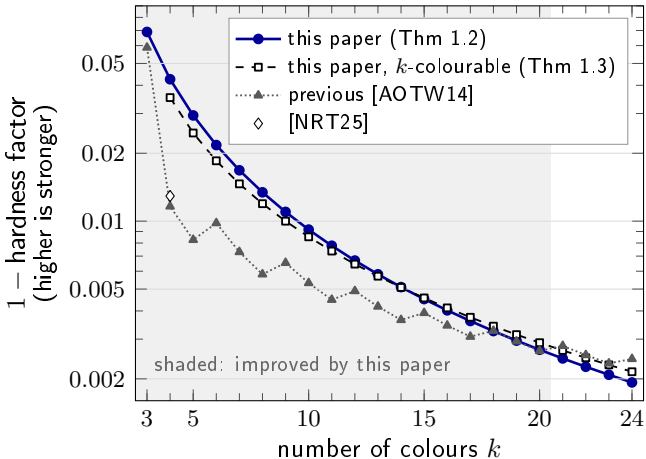
$k = 4$	$k = 5$
$45/47 \approx 0.957$	$33/34 \approx 0.971$
was 0.987 [NRT25]	was $120/121$ [AOTW14]

Theorem 1.3 (k-colourable graphs)

NP-hard to distinguish: k-colourable, or every k-colouring cuts at most

$k = 4$	$k = 5$
$82/85 + \varepsilon \approx 0.965$	$119/122 + \varepsilon \approx 0.975$
stated before: $85/86$	stated before: $120/121$

All k at a glance, and one limitation



- **improved** for every k from 3 to 20
- $k \geq 21$: previous bound [AOTW14] remains best

Limitation

No perfect completeness: on 3-colourable graphs, $16/17$ [AOTW14] is still the best bound.

Three ideas

1. **The negation trick** $\longrightarrow$ **$27/29$ for Max-3-Cut**

   pair every assignment X with $-X$; proof in a few lines (Section 3)

2. **A soundness criterion** for gadget reductions $\longrightarrow$ **all k**

   from Håstad's equations; used for more than three colours (Sections 4–5)

3. **A barrier** $\longrightarrow$ **$27/29$ is optimal for this method**

   under the criterion, gadgets of any size cannot do better (Section 7)

At the end: how these results were obtained — with extensive use of an AI system

Starting point: Håstad's theorem

Linear equations over $\mathbb{Z}_3$ with three variables:

$$x_1 + x_2 + x_3 = t$$

$t \in \{0, 1, 2\}$ is the **type** of the equation

example

$u_1 + w_1 + w_2 = 0$	type 0
$u_1 + w_2 + w_3 = 1$	type 1
$u_2 + w_3 + w_4 = 2$	type 2

Theorem 2.1 [Hås01]

For every $\varepsilon > 0$ it is NP-hard to distinguish

YES some assignment satisfies $\geq 1 - \varepsilon$ of the equations

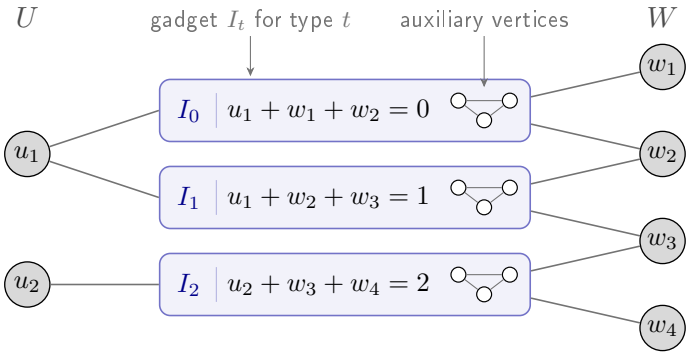
NO every assignment satisfies $\leq \frac{1}{3} + \varepsilon$ of the equations

random assignment: $\frac{1}{3}$

We use two features of these instances:

- **role separation**: one variable from U, two from W (U, W disjoint)
- **balance**: each type has total weight $\frac{1}{3}$

Gadget reductions



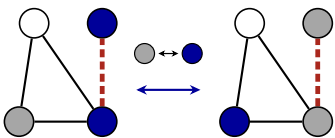
- Variable → **one vertex**; its colour = its value
- Equation → a **gadget**: a small graph with new auxiliary vertices
- Equation satisfied ⇒ the gadget can cut more

almost satisfiable system ⟶ **large cut**

far from satisfiable ⟶ **every colouring cuts less** [TSSW00]

The obstacle: colours have no names

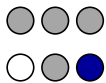
Rename the colours:



exactly the **same cut**

Type 0 is fine:

$x_1 + x_2 + x_3 = 0 \iff$



all equal, or

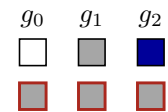
all distinct

unchanged when the colours are renamed

Types 1 and 2 are not:

which colour is "1"?

$\Rightarrow$ global **reference vertices**

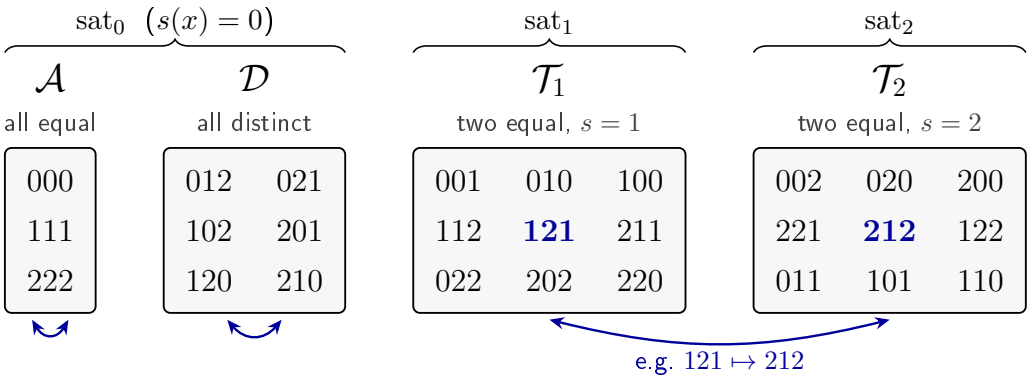


The colour-reference problem

Guarding against cheating makes the gadgets for $t \neq 0$ **large and inefficient**: for $k = 3$, the gadgets of [NRT25] give only $55/57 \approx 0.965$, weaker than $16/17 \approx 0.941$.

The 27 colour patterns of an equation

values $x = x_1x_2x_3 \in \mathbb{Z}_3^3$ of the three variables; $s = x_1 + x_2 + x_3$



Negation $x \mapsto -x$ (swap the values 1 and 2): $\mathcal{A} \to \mathcal{A}$, $\mathcal{D} \to \mathcal{D}$, $\mathcal{T}_1 \leftrightarrow \mathcal{T}_2$

The negation trick

Observation

If X is an assignment, so is $-X$, and

$$-X \text{ satisfies } x_1 + x_2 + x_3 = t \iff X \text{ satisfies } x_1 + x_2 + x_3 = -t$$

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NO case: every assignment X satisfies a fraction $\mathrm{val}(X) \leq \frac{1}{3} + \varepsilon$ of the equations.

Random equation: t = its type, s = the sum of its variables under X.

$$\Pr[s = t] + \Pr[s = -t] = \mathrm{val}(X) + \mathrm{val}(-X) \leq \frac{2}{3} + 2\varepsilon$$

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Types $t = 1$ and 2:

"$s = t$ or $s = -t$" $\iff$ $s \neq 0$ $\iff$ **exactly two values are equal** 

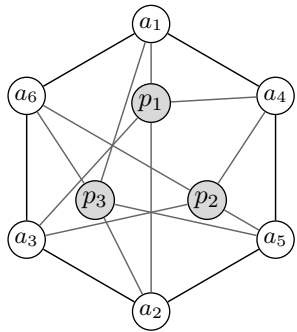
invariant under renaming the colours $\longrightarrow$ no reference vertices needed

Two gadgets

E

type 0

[NRT25]



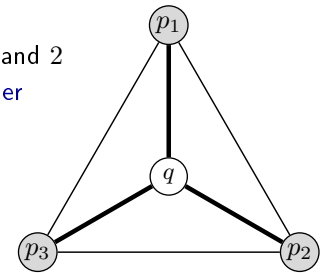
variable vertices

auxiliary vertices

N

types 1 and 2

this paper



6 auxiliary vertices, 18 unit edges

triangle + vertex q; thick edges: weight 3

max cut $= 18$ **if** $s = 0$, else 16 (Lemma 3.1)

max cut $= 11$ **if** $s \neq 0$, else 9

$s = x_1 + x_2 + x_3$: $s = 0$: all equal or all distinct; $s \neq 0$: exactly two equal

Proof of Theorem 1.1 on one slide

Gadgets: $2E$ for type 0, N for types 1 and 2 total edge weight: $\frac{1}{3}(36+12+12)=20$

$V_t(x)$: max cut of the type-t gadget on pattern x

$$V_t(x) \;=\; \beta_t + 2\big([s=t]+[s=-t]\big), \qquad \beta_0=32,\;\; \beta_1=\beta_2=9$$

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YES case: an assignment that satisfies $\geq 1-\varepsilon$ gives a colouring with

$$\text{cut} \;\geq\; \tfrac{1}{3}(36+11+11) - O(\varepsilon) \;=\; \tfrac{58}{3} - O(\varepsilon)$$

$\approx \mathbf{\frac{29}{30}}$ of the weight

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$\approx \mathbf{\frac{29}{30}}$ of the weight

NO case: any colouring gives an assignment σ, and

$$\text{cut} \;\leq\; \tfrac{50}{3} + 2\big(\text{val}(\sigma)+\text{val}(-\sigma)\big) \;\leq\; \tfrac{50}{3}+\tfrac{4}{3}+4\varepsilon \;=\; 18+4\varepsilon$$

$\approx \mathbf{\frac{9}{10}}$ of the weight

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Gadgets: $2E$ for type 0, N for types 1 and 2 total edge weight: $\frac{1}{3}(36+12+12) = 20$

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$\approx \mathbf{\frac{9}{10}}$ of the weight

$$\frac{9/10}{29/30} = \mathbf{\frac{27}{29}}$$

Two remarks

1 Judged gadget by gadget, N is useless

(Remark 3.3)

As a type-1 gadget, it cannot tell $s = 1$ from $s = 2$.

The gain comes from the analysis: pairing σ with $-\sigma$.

	$\mathcal{A}$	$\mathcal{D}$	$\mathcal{T}_1$	$\mathcal{T}_2$
V_N	9	9	11	11
			$s = 1$	$s = 2$

same value

2 The argument is robust

(Remark 3.2)

It needs neither role separation nor, essentially, balance.

if type 0 has weight λ_0, the ratio is

$$\frac{23\lambda_0 + \frac{31}{3}}{25\lambda_0 + 11} \leq \frac{31}{33} < \frac{16}{17}$$

More colours: a general soundness criterion

$k \geq 4$: satisfying colourings use only 3 of the k colours $\Rightarrow$ **global vertices** needed again

Theorem 4.1 (soundness criterion)

Gadgets I_0, I_1, I_2 for Max-k-Cut, one per type (each symmetric in the 2nd and 3rd variable). Then, for every $\delta > 0$, Max-k-Cut is NP-hard to approximate within $\rho + \delta$, where

$$\rho = \frac{\max\limits_{x,y} \sum\limits_{t} V_t(x,y)}{\sum\limits_{t} C_t}$$

← one colour pattern (x, y), all three gadgets

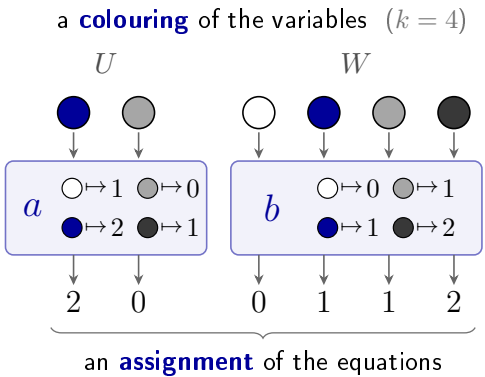
$V_t(x,y)$ best cut of the type-t gadget when its variables have colours x, the globals colours y

C_t its least value on satisfying colourings (globals coloured canonically)

A cheating colouring cannot do better than to use

one colour pattern on all equations

Why the criterion holds



- Colouring $\longrightarrow$ distribution $P(t, x)$ of (type, colour pattern)
- Any maps $a, b\colon [k] \to \mathbb{Z}_3$ **decode** it into an assignment
- **NO case:** it satisfies $\leq \frac{1}{3} + \varepsilon$
  $\Rightarrow$ every **test** (a, b) sees t as nearly uniform and independent of x
- **Spanning lemma**: these tests detect every dependence that can affect the cut
  short Fourier argument (Lemma 4.2)

Needs role separation: U and W are decoded by independent maps a and b

Tight: no black-box use of Håstad's theorem can improve the numerator of ρ (Proposition 4.3)

A uniform family for $k = 3 + L$ colours

Criterion $\Rightarrow$ gadget design is **one linear program** — solved for $k = 4, 5$, simplified by hand

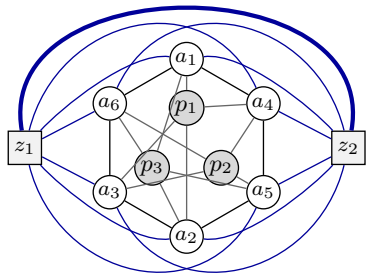
A_L

type 0

$= E +$ the z_i

weight 3

weight 1

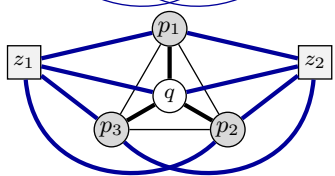


N_L

types 1 and 2

$= N +$ the z_i

drawn for $L = 2$



Add L **universal vertices** $z_1, \ldots, z_L$ (squares): globals for the extra colours

The family $(2A_L,\ N_L,\ N_L)$ gives

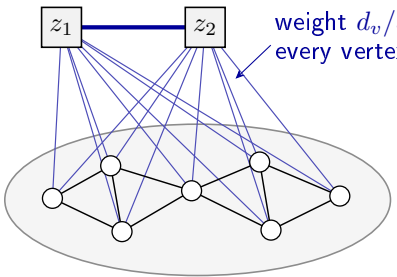
$$\rho = 1 - 2/\Gamma_L$$
$$\Gamma_L = 29 + 18L + 3\binom{L}{2}$$

(Lemma 5.1, by hand)

k	3	4	5	6	7
factor	$\frac{27}{29}$	$\frac{45}{47}$	$\frac{33}{34}$	$\frac{45}{46}$	$\frac{117}{119}$

Universal vertices and perfect completeness

A classical reduction [KKLP97, GS13]



G: hard instance for 3 colours

The new vertices z_i are **forced to take the extra colours**

(Lemma 6.1)

- **Preserves perfect completeness**
- Stated constants: applied on top of a weaker intermediate bound
- Applied directly to $\frac{16}{17}$ [AOTW14]:

$$1 - \frac{3}{51 + 34L + 3\binom{L}{2}} \qquad k = 3 + L$$

(Theorem 1.3)

4-colourable	$82/85$	stated before: $85/86$
5-colourable	$119/122$	stated before: $120/121$

Improves the stated bounds for $4 \le k \le 20$

Applied to Theorem 1.1: only $47/49$ for $k = 4$

A barrier at $27/29$

Theorem 7.1 (computer-assisted)

For **every** family (I_0, I_1, I_2) of Max-3-Cut gadgets — any number of auxiliary vertices, with or without global vertices:

$$\max_{x \in \mathbb{Z}_3^3} S_{y^\circ}(x) \quad \geq \quad \frac{\mathbf{27}}{\mathbf{29}} \sum_t C_t$$

$S_{y^\circ}(x) = \sum_t V_t(x, y^\circ)$: all three gadgets on one colour pattern x, with canonical globals

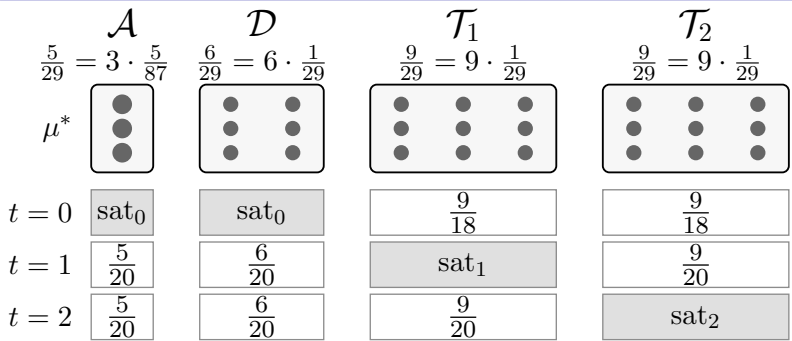
C_t: completeness of the type-t gadget (its least value on satisfying colourings)

$\Longrightarrow$ For Max-3-Cut, the criterion cannot give a factor below $27/29$: **the simple gadgets E and N are optimal**

$\Longrightarrow$ With tightness (Proposition 4.3): $27/29$ is **the limit for gadget reductions** from Håstad's equations that use his theorem as a black box

Proof: LP duality, in the spirit of [TSSW00, HHM+17]

The proof: one distribution that is bad for every gadget



μ^*: **uniform** within each class, masses

$$\mathbf{5 : 6 : 9 : 9}$$

$$\mathbb{E}_{\mu^*} V_t \,/\, C_t \;\geq$$

$$\frac{11}{29} + \frac{18}{29} \cdot \mathbf{\frac{8}{9}} = \mathbf{\frac{27}{29}}$$

$$\frac{9}{29} + \frac{20}{29} \cdot \mathbf{\frac{9}{10}} = \mathbf{\frac{27}{29}}$$

$$\frac{9}{29} + \frac{20}{29} \cdot \mathbf{\frac{9}{10}} = \mathbf{\frac{27}{29}}$$

Shaded cells

$V_t \geq C_t$ on sat_t, by definition

Other cells (μ^* given $x \notin \mathrm{sat}_t$)

an **LP-dual certificate** guarantees, on average, $\mathbf{\frac{8}{9}}\, C_0$ (type 0) and $\mathbf{\frac{9}{10}}\, C_t$ (types 1, 2) (Lemma 7.2)

Exact rational certificates

78, 162, 162 rows; verified in exact arithmetic by **two independent programs**

How these results were obtained

This research was carried out with **extensive use of an AI system** (Claude, Anthropic), under the direction of the author. Details: Appendix B and the AI methodology statement

The AI system

- proposed the main ideas
- wrote and ran the search and verification programs
- searched the literature
- drafted the paper

How the claims were checked

- **Independent AI agents** tried to refute every main claim: re-read the primary sources, re-derived the proofs, re-implemented the computations
- Every gadget value and certificate: verified by ≥ 2 **independent programs**, in exact arithmetic
- Theorems 1.1–1.3 have **short proofs by hand**; only Theorem 7.1 is computer-assisted

The author verified all content and is responsible for it.

Code, gadget families, certificates: `github.com/ziangni-sys/max-3-cut`

Open problems

1. **Perfect completeness.** Beat $16/17$ [AOTW14] on 3-colourable graphs?

   combine the negation trick with a source that has perfect completeness

2. **Other sources.** Below $27/29$: change the source, reduction, or analysis

   candidates: Chan's predicates [Cha16], where the negation trick applies as well

3. **$k \geq 4$.** Are $45/47$ and $33/34$ optimal for the criterion?

   so far, no barrier certificate

4. **Large k.** Can the criterion improve the constant in $1 - \Theta(1/k)$?

   the family of this paper gives only $1 - \Theta(1/k^2)$

... and $k = 3$: 0.836 (optimal under the Unique Games Conjecture) vs. 0.931 (NP-hard)

Summary

Max-3-Cut is NP-hard to approximate within $\mathbf{27/29} + \varepsilon \approx 0.931$

was $16/17 \approx 0.941$ since 2012 [AOTW14]

Negation trick: pair every assignment X with $-X$

$\longrightarrow$ colour-symmetric predicates, tiny gadgets

A tight **soundness criterion**, and a matching **barrier** at $27/29$

$\longrightarrow$ more colours; with Theorem 1.3: better bounds for every k from 3 to 20

Thank you! code and certificates: `github.com/ziangni-sys/max-3-cut`